

REVIEW

• FFT

e.g.
$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}_{4 \times 1}$$
 1. Find the 4th primitive root of unity.
 $\{\pm i\}$

2. Construct matrix
 ij^{th} entry = ω^{ij}

$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} i^0 & i^0 & i^0 & i^0 \\ i^0 & i & i^2 & i^3 \\ i^0 & i^2 & i^4 & i^6 \\ i^0 & i^3 & i^6 & i^9 \end{pmatrix} \end{matrix} \equiv \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

SHOULD BE SYMMETRIC

3. Compute $(b_0 \ b_1 \ b_2 \ b_3)$

$$\begin{aligned} b_0 &= a_0 + a_1 + a_2 + a_3 \\ b_1 &= a_0 + a_1 i - a_2 - a_3 i \\ b_2 &= a_0 - a_1 + a_2 - a_3 \\ b_3 &= a_0 - a_1 i - a_2 + a_3 i \end{aligned}$$

$$\begin{aligned} b_i &= \sum_{j=0}^{n-1} \omega^{ij} \cdot a_j = \left(\sum_{j=0}^{n/2-1} \omega^{ij} \cdot a_j \right) + \left(\sum_{j=n/2}^{n-1} \omega^{ij} \cdot a_j \right) \\ &= \sum_{j=0}^{n/2-1} \omega^{ij} \cdot a_j + \left(\omega^{\frac{n}{2}i} \left(\sum_{j=0}^{n/2-1} \omega^{ij} \cdot a_{j+\frac{n}{2}} \right) \right) \\ &\quad \downarrow j \leftarrow j - \frac{n}{2} \end{aligned}$$

Say $i = 2p$

$$\Rightarrow \sum_{j=0}^{n/2-1} \omega^{ij} \cdot a_j + \left[\omega^{\frac{n}{2} \cdot 2p} \left(\sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot a_{j+\frac{n}{2}} \right) \right] = \sum_{j=0}^{n/2-1} \left[(\omega^2)^{pj} \cdot (a_j + a_{j+\frac{n}{2}}) \right]$$

$$\therefore b_0 = \sum_{j=0}^1 (\omega^2)^{0 \cdot j} (a_j + a_{j+2}) = (a_0 + a_2) + (a_1 + a_3)$$

$$\begin{aligned} b_2 &= \sum_{j=0}^1 (\omega^2)^{1 \cdot j} (a_j + a_{j+2}) = (\omega^2)^{1 \cdot 0} \cdot (a_0 + a_2) + (\omega^2)^{1 \cdot 1} \cdot (a_1 + a_3) \\ &= (a_0 + a_2) - (a_1 + a_3) \\ &= a_0 - a_1 + a_2 - a_3 \end{aligned}$$

Say $i = 2p+1$

$$\begin{aligned} &\Rightarrow \left(\sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot \omega^j \cdot a_j \right) - \left(\sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot \omega^j \cdot a_{j+\frac{n}{2}} \right) \\ &= \sum_{j=0}^{n/2-1} (\omega^2)^{pj} \cdot (a_j - a_{j+\frac{n}{2}}) \cdot \omega^j \end{aligned}$$

$$\begin{aligned} \therefore b_1 &= \sum_{j=0}^1 (\omega^2)^{0 \cdot j} \cdot (a_j - a_{j+\frac{n}{2}}) \cdot \omega^j = (a_0 - a_2) \cdot \omega^0 + (a_1 - a_3) \cdot i \\ &= a_0 + a_1 i - a_2 - a_3 i \end{aligned}$$

$$\begin{aligned} b_3 &= \sum_{j=0}^1 (\omega^2)^{1 \cdot j} \cdot (a_j - a_{j+\frac{n}{2}}) \cdot \omega^j = 1 \cdot (a_0 - a_2) \cdot \omega^0 + (\omega^2) \cdot (a_1 - a_3) \cdot \omega \\ &= (a_0 - a_2) + \omega^3 (a_1 - a_3) \\ &= a_0 - a_2 - a_1 i + a_3 i \end{aligned}$$

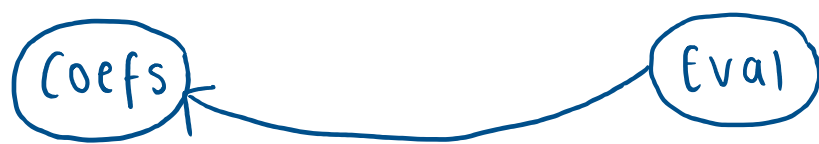
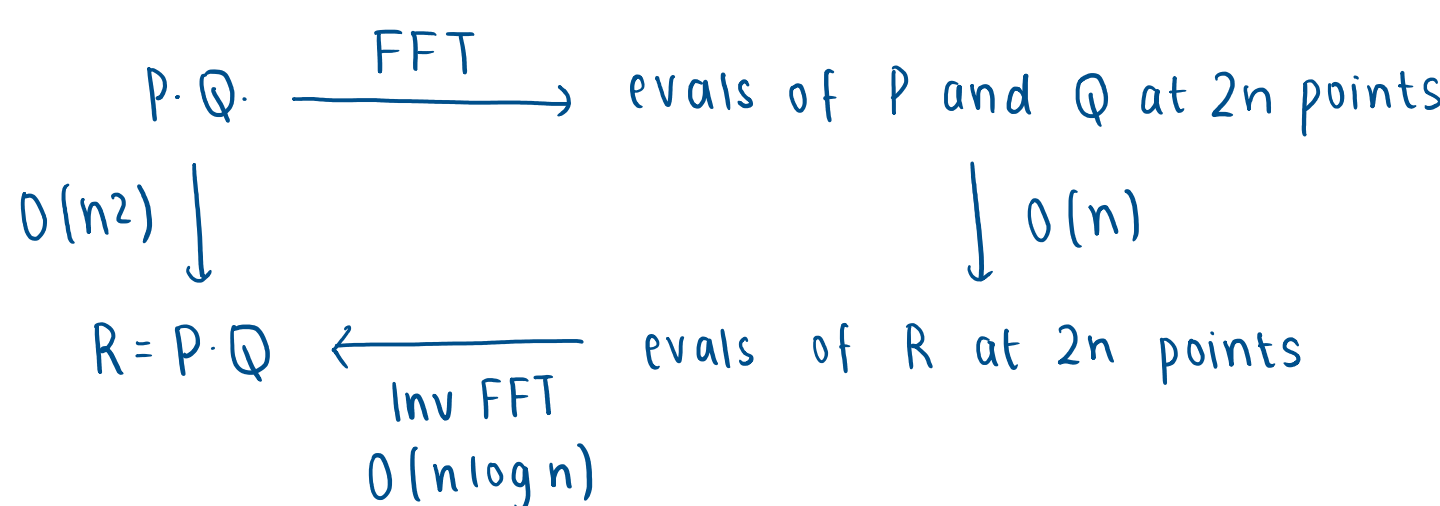
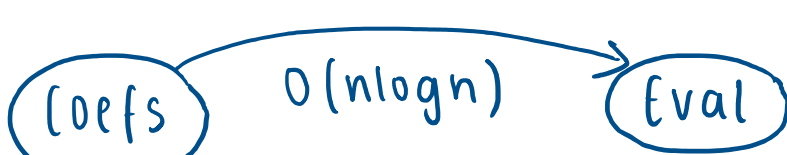
• DFT of dim 2

$$\begin{matrix} 0 & \begin{matrix} 0 & 1 \end{matrix} \\ 1 & \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{matrix} \quad \omega = 2^{nd} \text{ root of unity}$$
$$V_e = \begin{pmatrix} a_0 + a_{n/2} \\ a_1 + a_{n/2+1} \\ \dots \\ a_{n/2-1} + a_n \end{pmatrix}, \quad V_o = \begin{pmatrix} (a_0 - a_{n/2}) \omega^0 \\ (a_1 - a_{n/2+1}) \cdot \omega^1 \\ \dots \\ (a_{n/2-1} - a_n) \omega^{n/2-1} \end{pmatrix}$$

$$V_e = \begin{pmatrix} a_0 + a_2 \\ a_1 + a_3 \end{pmatrix} \quad V_o = \begin{pmatrix} a_0 - a_2 \\ (a_1 - a_3) i \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a_0 + a_2 \\ a_1 + a_3 \end{pmatrix} = \begin{pmatrix} a_0 + a_1 + a_2 + a_3 \rightarrow b_0 \\ a_0 - a_1 + a_2 - a_3 \rightarrow b_2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a_0 - a_2 \\ (a_1 - a_3) i \end{pmatrix} = \begin{pmatrix} a_0 - a_2 + a_1 i - a_3 i \rightarrow b_1 \\ a_0 - a_2 - a_1 i + a_3 i \rightarrow b_3 \end{pmatrix}$$



• Inverse DFT

$$\omega^{ij} = \frac{\omega^{-ij}}{n} = \frac{\omega^{n-ij}}{n}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \times \underbrace{\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{\text{Inv DFT}_2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} b_i &= \sum_{j=0}^{n-1} \frac{\omega^{n-ij}}{n} \cdot a_j \\ b_i &= \sum_{j=0}^{n-1} \frac{\omega^{n-ij}}{n} \cdot a_j = \left(\sum_{j=0}^{n/2-1} \left(\frac{\omega^{n-ij}}{n} \right) \cdot a_j \right) + \left(\sum_{j=n/2}^{n-1} \left(\frac{\omega^{-ij}}{n} \right) \cdot a_j \right) \\ &\quad \downarrow 0 \quad \downarrow j+\frac{n}{2} \\ &= \left(\sum_{j=0}^{n/2-1} \left(\frac{\omega^{-ij}}{n} \right) a_j \right) + (\omega^{-\frac{n}{2}i}) \left(\sum_{j=0}^{n/2-1} \frac{\omega^{-ij}}{n} \cdot a_{j+\frac{n}{2}} \right) \end{aligned}$$

Say $i = 2p$

$$\begin{aligned} &\Rightarrow \left(\sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{n} \cdot a_j \right) + (\omega^{-\frac{n}{2} \cdot 2p}) \left(\sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{n} \cdot a_{j+\frac{n}{2}} \right) \\ &= \sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{n} \cdot (a_j + a_{j+\frac{n}{2}}) \\ &= \sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{(n/2)} \cdot \frac{(a_j + a_{j+n/2})}{2} \end{aligned}$$

$$V_e = \begin{pmatrix} (a_0 + a_{n/2})/2 \\ (a_1 + a_{n/2+1})/2 \\ \vdots \\ (a_{n/2-1} + a_{n-1})/2 \end{pmatrix}$$

Say $i = 2p+1$

$$\begin{aligned} &\Rightarrow \left(\sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{n} \cdot a_j \cdot \omega^j \right) + \omega^{-\frac{n}{2} \cdot (2p+1)} \left(\sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{n} \cdot a_{j+\frac{n}{2}} \cdot \omega^j \right) \\ &= \sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{n} \cdot (a_j - a_{j+\frac{n}{2}}) \cdot \omega^j \\ &= \sum_{j=0}^{n/2-1} \frac{(\omega^2)^{-pj}}{(n/2)} \cdot \frac{(a_j - a_{j+n/2})}{2 \omega^j} \end{aligned}$$

$$V_o = \begin{pmatrix} (a_0 - a_{n/2}) \cdot 1/2 \omega^0 \\ (a_1 - a_{n/2+1}) \cdot 1/2 \omega^1 \\ \vdots \\ (a_{n/2-1} - a_{n-1}) \cdot 1/2 \omega^{n/2-1} \end{pmatrix}$$